Joseph Lavond

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Education

University of North Carolina — Chapel Hill, NC
Ph.D. Statistics and Operations Research — *Aug. 2020 – May 2025*
M.S. Statistics and Data Science — *Aug. 2020 – May 2024*

California Polytechnic State University — San Luis Obispo
B.S. Statistics — *Sep. 2016 – Mar. 2020*

Professional Experience

Data Scientist — June 2025 – Present
Red Ventures — *Fort Mill, SC*

- Authored the LLM cost-optimization roadmap for Kamino, RV's internal AI agent platform (15 business-unit installations, 120+ agents), and delivered its measurement and prompt-caching phases, cutting cost per model call ∼30% (∼$65K/yr at current volume)
- Built production trace monitoring for Kamino agents that prioritizes conversations by implicit signals (trace errors) and explicit feedback (chat reactions), classifies failures, and turns them into eval cases and recommendations A/B validated against stored tests; live in 3 business units
- Built and productionized ReplayOPE, an offline policy evaluation platform for contextual bandits (FastAPI, Databricks Apps, React UI, MCP); challenger configurations beat production baselines in 67% of 54 analyses (3.2% median lift), and a change-tracking pipeline links recommendations to production changes across 18 agents
- Developed advanced evaluation methods including combinatorially purged cross-validation, fold-based evaluation, warm starts, delayed rewards, streaming execution, concurrent runs, eligibility checks, and statistically rigorous policy comparison
- Hardened offline RL training infrastructure: fixed a 3-year missing-values bug and a 2-year policy-importance bug, and productionized CatBoost, ordinal encoding, and recency-based sample weighting, with Replay-estimated revenue lifts of up to 10% on several agents
- Onboarded 20+ analysts and SMEs to ReplayOPE through live demos, docs, and API/MCP access, producing 18+ experiment runs across 6 clients within 3 weeks of the UI launch
- Architected a Claude Code skills marketplace with LLM security gating and SHA-pinned releases, adopted by the security org as the company-wide gate for approved skills
- Managed and mentored a summer intern, whose main-branch commits rose from 2 to 44 after a midsummer coaching change; recommended for a full-time return offer

Graduate Data Science Intern — June 2024 – Aug. 2024
Red Ventures — *Fort Mill, SC*

- Implemented offline model-based Q-learning for website decisioning, demonstrating a typical 5% revenue-per-session lift and estimated $5M annual revenue opportunity
- Developed 6K+ lines of production code for a Markov Decision Process simulation environment and built a historical-production-data pipeline for larger, more complex simulation experiments

Ph.D. Intern — June 2023 – Aug. 2023
Cisco — *San Jose, CA*

- Developed scalable differentially private generative-model training methods and evaluated GAN mode-collapse mitigation techniques to improve private model training reliability

Graduate Info Technology Intern — June 2022 – Aug. 2022
Elevance Health — *Indianapolis, IN*

- Built a PyTorch semi-supervised Bayesian anomaly-detection model on AWS to identify exaggerated provider claims
- Created a PySpark SQL modeling pipeline for 60M-row claims data in Jupyter on Kubeflow; received end-of-summer presentation award

Teaching & Research Experience

Graduate Teaching Fellow Spring 2022 & Fall 2024

University of North Carolina *Chapel Hill, NC*

- Taught introductory statistics and data science to 100+ undergraduates across multiple semesters
- Designed student research projects on data storytelling and statistical bias to develop applied reasoning and communication skills

NSF-Funded Research Training Group Aug. 2022 – May 2025

University of North Carolina *Chapel Hill, NC*

- Selected for interdisciplinary training supported by a $2M grant on the theory and application of networks

Selected Publications

- Joseph Lavond, Yuexin Cheng, Chudi Zhong, and Yao Li. Partition-based backdoor detection and purification from poisoned data. 2026. Under Review
- Joseph Lavond, Minhao Cheng, and Yao Li. Feddecay: Balancing model performance and rapid personalization in federated learning with learning rate scheduling. *Journal of Computational and Graph Statistics*, 2026. Accepted
- Joseph Lavond. *Advancing Model Security, Data Privacy, and Performance for Widespread Adoption of Trustworthy Artificial Intelligence*. PhD thesis, University of North Carolina at Chapel Hill, 2025
- Joseph Lavond, Minhao Cheng, and Yao Li. Trusted aggregation (TAG): Backdoor defense in federated learning. *Transactions on Machine Learning Research*, 2024

Selected Projects

Statistical Consultant, University of North Carolina Aug. 2022 – Dec. 2023

- Used R to analyze longitudinal behavioral and operational data, identifying early autism-risk markers and key drivers of medical contract turnaround delays

Fatal Force, University of North Carolina Aug. 2020 – May 2021

- Engineered features from FBI crime reports and Census data and evaluated statistical models predicting police shooting fatalities, improving test RMSE by 3%

Frost Summer Research Program, Cal Poly May 2019 – Sep. 2019

- Built an R Shiny survival-analysis app for cancer prognosis using KM and Cox models and processed SEER data covering 20M cancer cases

Skills

ML & Decision Systems: Contextual Bandits, Offline Policy Evaluation (OPE), Reinforcement Learning, Policy Evaluation, LightGBM, CatBoost, Ordinal Encoding, Sample Weighting, Historical Importance, Ensemble Methods, scikit-learn

Statistics & Deep Learning: Bayesian Inference, Experiment Design, Cross-Validation, Purged Cross-Validation, A/B Testing, Statistical Testing, Policy Comparison, PyTorch, TensorFlow, Keras

LLM & Agent Systems: LLM Evaluation, Trace Monitoring, Prompt Caching, Cost Attribution, MCP, Claude Code Plugins, AWS Bedrock

ML Systems & Infrastructure: FastAPI, TypeScript/React, Databricks Apps/Jobs, GitHub Actions, Terraform, New Relic, CI/CD, Model Deployment, API Design

Programming & Data: Python, SQL, R, Pandas, NumPy, PySpark, SciPy, Tidyverse, Feature Engineering, Data Pipelines, Data Quality & Validation

Analytics, Visualization & Tools: Clustering, Segmentation, Dashboarding, Matplotlib, Seaborn, ggplot2, Git, Linux